Shape vs. Appearance: Comparing DINOv2 and YOLOv8 on Original and Hollow Contour Images

Abstract

We investigate how well vision models can classify images using only shape outlines (Canny edge contours) compared to full-color originals. We compare two architectures—DINOv2 (a self-supervised Vision Transformer) and YOLOv8n-cls (a lightweight CNN classifier)—on CIFAR-10 in both original and contour-only form. Both models achieve ∼96% on originals but diverge sharply on contours: YOLOv8n retains 70.4% while DINOv2 drops to 51.6%. We further evaluate both models on a custom Overwatch character recognition dataset (14 classes), where YOLOv8n achieves 65.8% and DINOv2 achieves 56.5%. Across experiments, fine-tuned CNNs consistently outperform frozen self-supervised features.

1 Introduction

A long-standing question in visual recognition is the relative importance of shape versus texture in how neural networks classify objects [1]. Human perception is strongly shape-biased, yet convolutional neural networks have been shown to rely more on local texture cues. Self-supervised Vision Transformers such as DINO [2] have been reported to develop more shape-aware representations, attending to object boundaries even without explicit supervision.

In this study, we test this claim empirically using two datasets and two contrasting architectures:

Datasets:

- **CIFAR-10**: A standard 10-class object recognition benchmark (50k train, 10k test; native 32×32). We evaluate on both original images and Canny edge contour versions.
- **Overwatch Frames**: A custom 14-class dataset of game character screenshots (1,309 train, 336 test; native 3420×2138), containing both normal and partially occluded views. We train directly on this dataset to evaluate character recognition.

Models:

- **DINOv2** (ViT-S/14): A frozen self-supervised Vision Transformer with a linear classification head trained on the target data.
- **YOLOv8n-cls**: A lightweight convolutional classifier, fully fine-tuned from pretrained ImageNet weights.

2 Method

2.1 CIFAR-10 Dataset Preparation

We use CIFAR-10 (50,000 training images, 10,000 test images across 10 classes). For each image, we generate a contour version by: (1) converting to grayscale, (2) applying Canny edge detection (thresholds: 100, 200), and (3) saving the binary edge map as a 3-channel PNG. All images are resized to 224×224. Figure 1 shows example pairs.

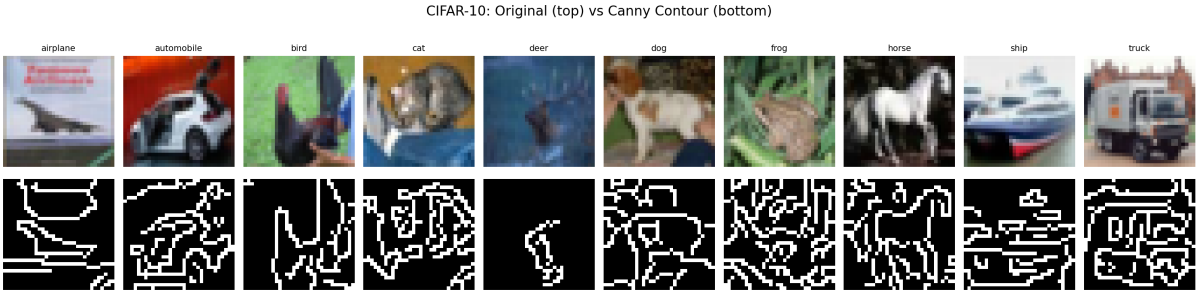

Figure 1: CIFAR-10 samples. Top: originals. Bottom: Canny edge contours.

2.2 Overwatch Dataset Preparation

The Overwatch dataset consists of in-game screenshots of 14 characters: Ashe, Bastion, Cassidy, Echo, Freja, Reaper, Sojourn, Soldier, Sombra, Symmetra, Torbjourn, Tracer, Venture, and Widowmaker. Each character has images from both `normal` (unobstructed) and `occluded` (partially hidden) viewpoints. We merge both conditions per character and split 80/20 into train/test sets, yielding approximately 93 training and 24 test images per class. All images are resized to 224×224. Figure 2 shows examples.

Figure 2: Overwatch character samples (resized to 224×224).

2.3 DINOv2 Setup

We use the `dinov2_vits14` model from the official DINOv2 release. The ViT-S/14 backbone is frozen, and we train only a linear classifier (384-dim $\rightarrow N$ classes) on top of the `[CLS]` token output. Training runs for 100 epochs with the Adam optimizer (lr $= 1 \times 10^{-3}$) and cosine annealing. Batch size is 256 for CIFAR-10 and 64 for Overwatch. Data augmentation (random horizontal flip, color jitter) is applied for Overwatch.

2.4 YOLOv8n-cls Setup

We use the Ultralytics YOLOv8n-cls model, pretrained on ImageNet. All layers are fine-tuned for 30 epochs with the AdamW optimizer (auto-tuned learning rate), image size 224×224. Batch size is 256 for CIFAR-10 and 64 for Overwatch.

2.5 Hardware

All experiments were run on a single NVIDIA RTX 6000 Ada Generation (48 GB) with PyTorch 2.6 and CUDA 12.4.

3 Results: CIFAR-10 (Original vs. Contour)

3.1 Overall Accuracy

Table 1: Test accuracy (%) on CIFAR-10.

Model	Original	Contour
DINOv2 (ViT-S/14, frozen + linear)	95.89	51.59
YOLOv8n-cls (fine-tuned)	96.02	70.38

Both models perform comparably on original images (∼96%). However, when texture and color are removed, YOLOv8n retains significantly more performance (70.4%) than DINOv2 (51.6%)—a gap of nearly 19 percentage points.

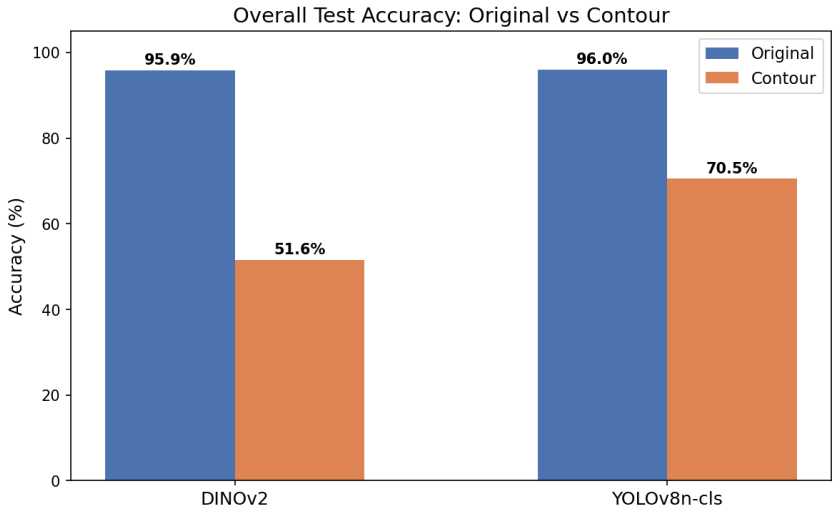

Figure 3: CIFAR-10: Overall accuracy comparison.

3.2 Per-Class Analysis

Figure 4 breaks down accuracy by class. On contour images, YOLO consistently outperforms DINO across all classes. Classes with distinctive shapes (*ship*, *airplane*, *truck*) retain relatively high accuracy for both models, while classes with less distinctive contours (*cat*, *deer*, *dog*) suffer the most.

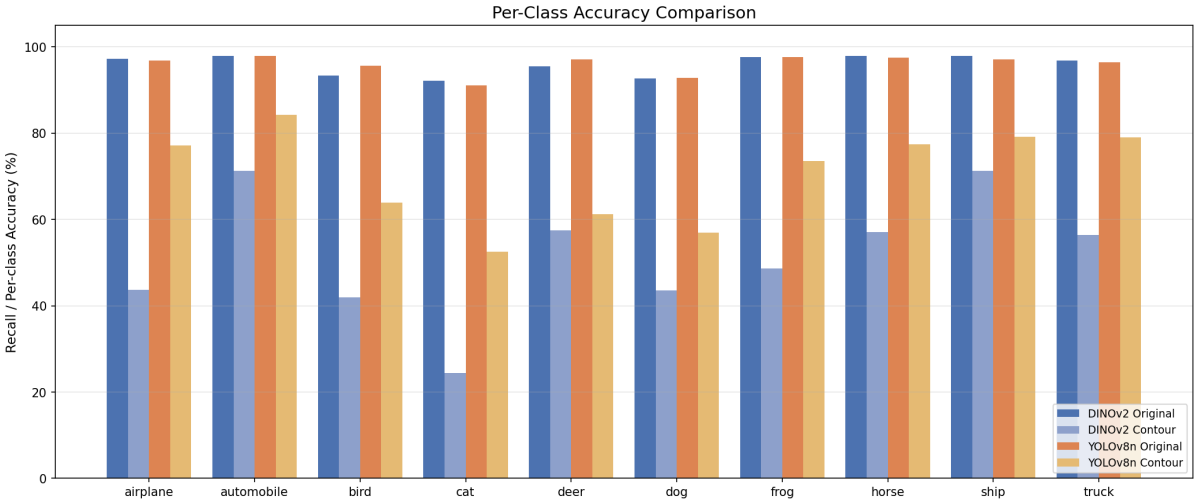

Figure 4: CIFAR-10: Per-class accuracy (recall).

3.3 Confusion Matrices

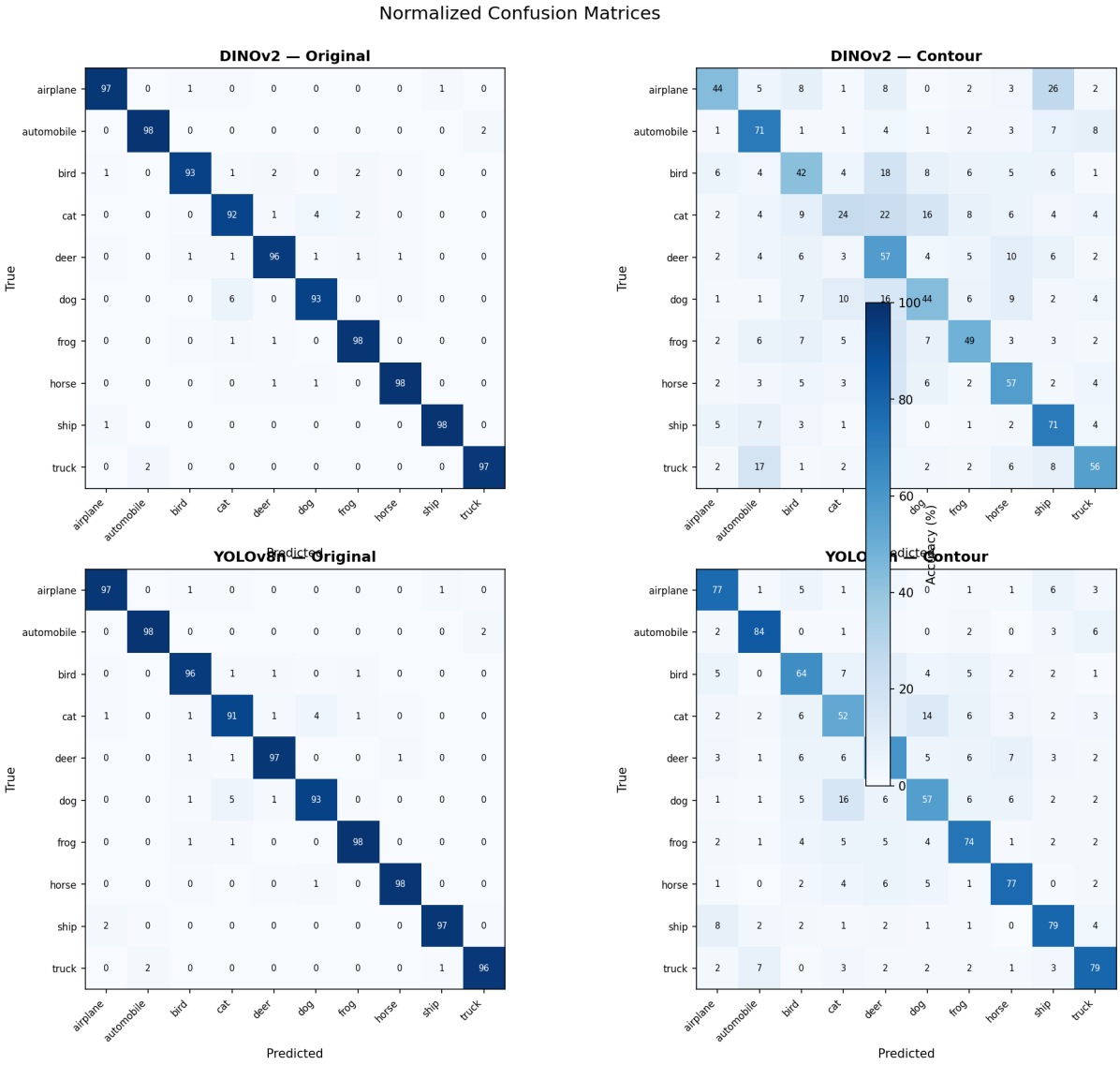

Figure 5: CIFAR-10: Normalized confusion matrices.

3.4 Training Curves

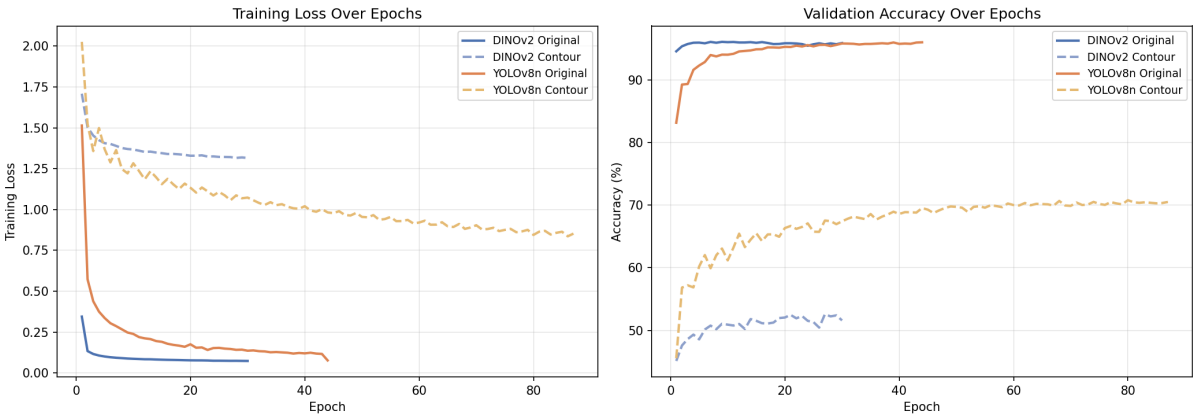

Figure 6: CIFAR-10: Training loss and validation accuracy over epochs.

4 Results: Overwatch Character Recognition

4.1 Overall Accuracy

Table 2: Test accuracy (%) on Overwatch character dataset (14 classes, random baseline = 7.1%).

Model	**Accuracy**
DINOv2 (ViT-S/14, frozen + linear)	56.55
YOLOv8n-cls (fine-tuned)	65.77

The Overwatch dataset is substantially harder than CIFAR-10: it has more classes (14 vs. 10), far fewer training examples (∼93 per class vs. 5,000), and the characters share similar in-game environments, lighting, and visual styles. Despite this, both models perform well above the 7.1% random baseline. YOLOv8n outperforms DINOv2 by 9.2 percentage points, consistent with the CIFAR-10 finding that full fine-tuning provides a significant advantage over a frozen backbone with a linear probe.

4.2 Per-Class Analysis

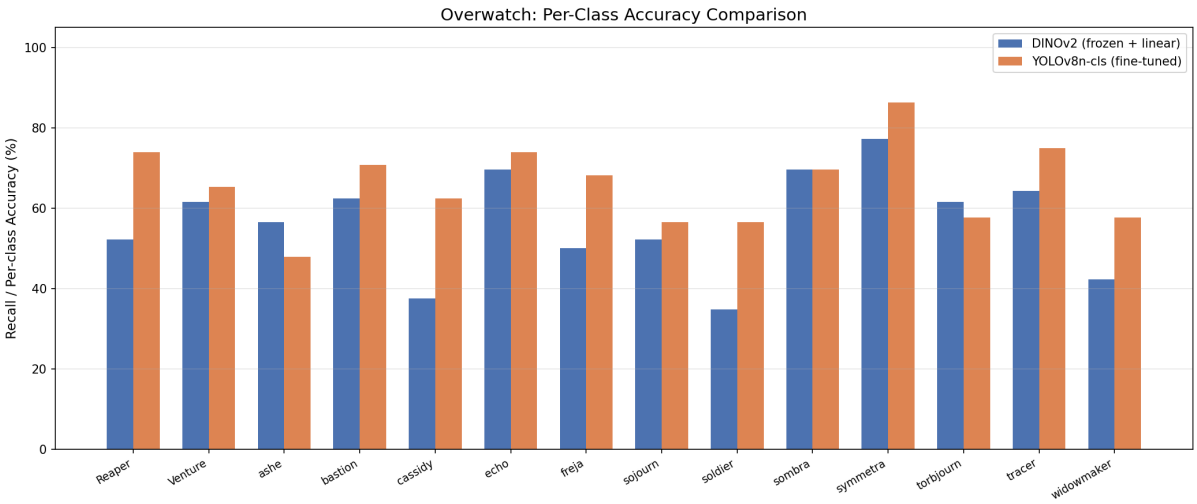

Figure 7: Overwatch: Per-class accuracy (recall).

Figure 7 shows that character recognition accuracy varies widely across classes. Characters with visually distinctive silhouettes or color schemes (e.g., Bastion, Tracer) tend to be easier, while characters that share similar body types or appear in similar environments are more frequently confused.

4.3 Confusion Matrices

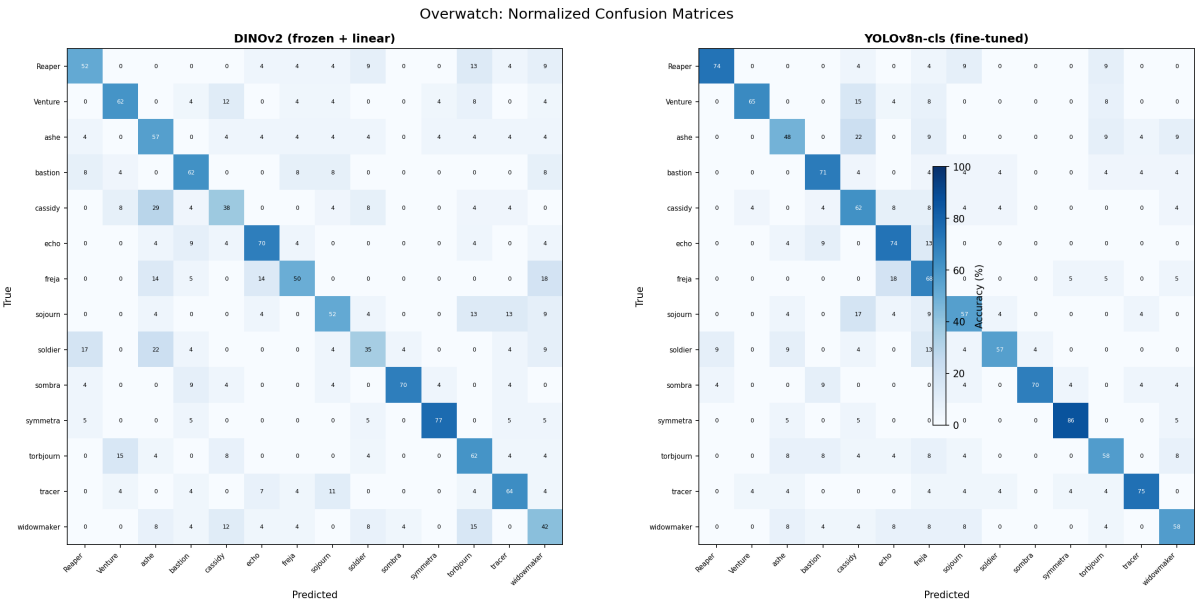

Figure 8: Overwatch: Normalized confusion matrices.

4.4 Training Curves

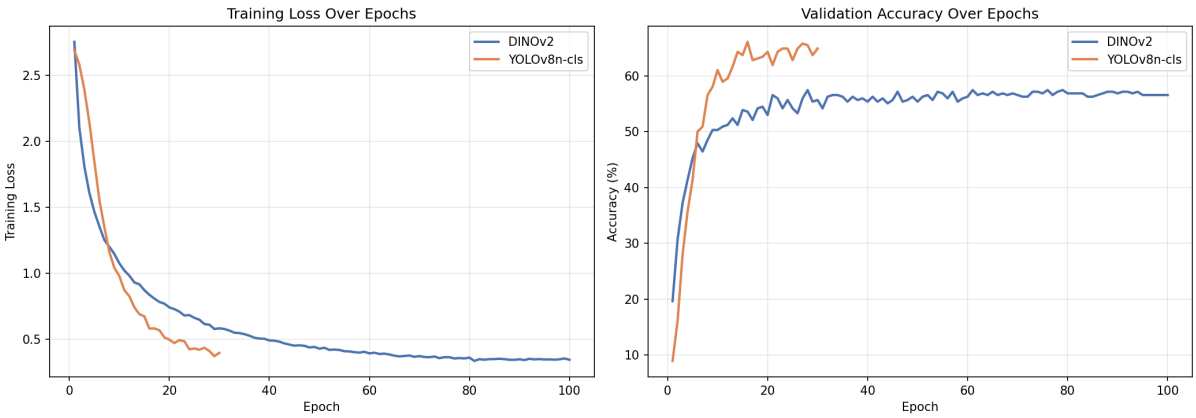

Figure 9: Overwatch: Training loss and validation accuracy over epochs.

5 Discussion

5.1 Fine-tuning vs. Frozen Features

Across both datasets, YOLOv8n-cls (fully fine-tuned) consistently outperforms DINOv2 (frozen backbone + linear probe). On CIFAR-10 originals the gap is small (<1%), but it widens on harder tasks: 19 points on CIFAR-10 contours and 9 points on Overwatch.

The key factor is *adaptation*. DINOv2's self-supervised features are powerful general-purpose representations, but a linear probe cannot reshape them for domains that differ substantially from natural images. Full fine-tuning, as in YOLOv8n, allows the model to adapt all layers—including low-level edge detectors and mid-level texture encoders—to the specific characteristics of the target data.

5.2 Shape Information in Contours

The CIFAR-10 contour experiments demonstrate that Canny edge outlines retain meaningful classification signal: YOLOv8n achieves 70.4% accuracy from shape alone. However, the ∼26% drop from originals confirms that texture and color remain important complementary cues.

5.3 Challenges of Game Character Recognition

The Overwatch dataset presents unique challenges:

- **Small training set**: ∼93 images per class limits what both models can learn.
- **Shared visual context**: All characters appear in similar game environments, so the model must rely on character-specific features rather than background cues.
- **Occlusion**: The inclusion of partially occluded views adds within-class variability and makes recognition harder.

Despite these challenges, YOLOv8n reaches 65.8%, demonstrating that fine-tuned models can learn character-discriminative features even from limited data.

6 Conclusion

We compared DINOv2 and YOLOv8n-cls across two tasks: CIFAR-10 classification (with original and contour images) and Overwatch character recognition. YOLOv8n consistently outperforms DINOv2 across all conditions. The CIFAR-10 contour experiments show that shape alone carries substantial but incomplete classification signal, with fine-tuned models much better at exploiting it. On the Overwatch dataset, YOLOv8n achieves 65.8% on a challenging 14-class problem with limited training data, outperforming DINOv2's 56.5%. The primary factor across all experiments is training strategy—full fine-tuning vs. frozen backbone—rather than architecture (CNN vs. ViT).

References

[1] R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, "ImageNet-trained CNNs are biased towards textures; increasing shape bias improves accuracy and robustness," in *Proc. ICLR*, 2019.

[2] M. Caron, H. Touvron, I. Misra, H. Jégou, J. Mairal, P. Bojanowski, and A. Joulin, "Emerging properties in self-supervised vision transformers," in *Proc. ICCV*, 2021.